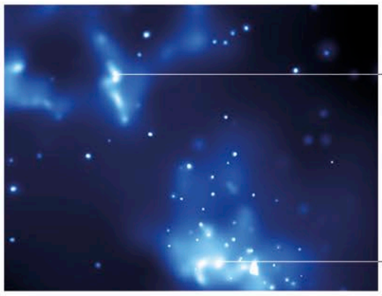
Ancient nucleus
Light from quasars is redshifted by huge amounts, revealing, due to the expansion of the Universe (see pp.70–71), that they are billions of light-years away and incredibly bright. As a result we are seeing them during a much earlier stage of cosmic evolution. Astronomers suspect that most galaxies go through a quasar phase early in their history.



△ **The active Milky Way**
Our own galaxy's supermassive black hole has long ago swept up the material from its immediate surroundings and become dormant, but objects such as stray asteroids still occasionally wander into its grasp. When this happens, matter is violently torn apart, resulting in intense bursts of radiation.